

# Beyond Static Imitation: Auditing LLM-Derived Traveler Responses for Transport Simulation

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## ABSTRACT

Transport scenario analysis requires traveler models that reproduce changes in choices, not only baseline predictions. For models distilled from large language models (LLMs), agreement with a Teacher does not establish agreement with travelers or preservation of responses in simulation. We develop a response-centric evaluation that links three comparisons: paired Teacher–Student contrasts, agreement with stated choices, and transmission through assignment and network execution. A persona-disjoint benchmark from one LLM Teacher contains 398 test response pairs across six synthetic personas, evaluated with three neural training seeds. Explicit response supervision reduces mean absolute probability–response error from 0.06930 to 0.06586; a multinomial logit Student reduces it further to 0.05644. Pairing the logit model with independent neural timing yields a departure error of 7.61 minutes against the Teacher. The model ranking changes when evaluated on stated-choice convenience samples of 332 Singapore and 321 Shanghai respondents. For poorer public-transport accessibility in Singapore, the Teacher-best Student predicts a 13.85-percentage-point increase in PT use, whereas respondents report a 24.40-point decrease. A 1,000-traveler Helsinki execution experiment further separates response preservation from routability: feasibility-constrained assignment removes unroutable PT assignments without reducing the response gap under a perceived-delay perturbation. These results establish distinct tests for compression fidelity, agreement with stated behavior and execution consistency. Model selection for transport simulation should evaluate the relevant intervention response at each level rather than rely on a single imitation or prediction score.

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## 1. Introduction

Transport scenario analysis asks how travel demand changes when conditions change. Matching a baseline mode share does not establish whether a model will reproduce the response to a delay, a fare increase or a longer walk to a stop. In agent-based simulation, such responses pass through assignment, routing and interaction with transport supply before becoming simulated demand (Horni et al., 2016; Axhausen, 2002). The relevant behavioral target is therefore a change between conditions, alongside the level of demand in each condition.

Large language models (LLMs) offer a flexible way to specify traveler behavior. Prompts can combine personal circumstances, trip attributes and contextual changes, supporting applications in travel prediction, route choice and agent-based modeling (Mo et al., 2026; Wang et al., 2025; Liu et al., 2025; Nie et al., 2025). Behavioral distillation moves repeated inference offline: an LLM Teacher supplies numerical judgments, and a compact Student uses these targets to make local predictions. This is a concrete instance of the hybrid LLM–conventional-model designs proposed for transport simulation (Liu et al., 2025). It reduces reliance on live LLM calls during deployment, but introduces a distinct validation problem: which properties of the Teacher survive compression, and which of those properties are useful for transport analysis?

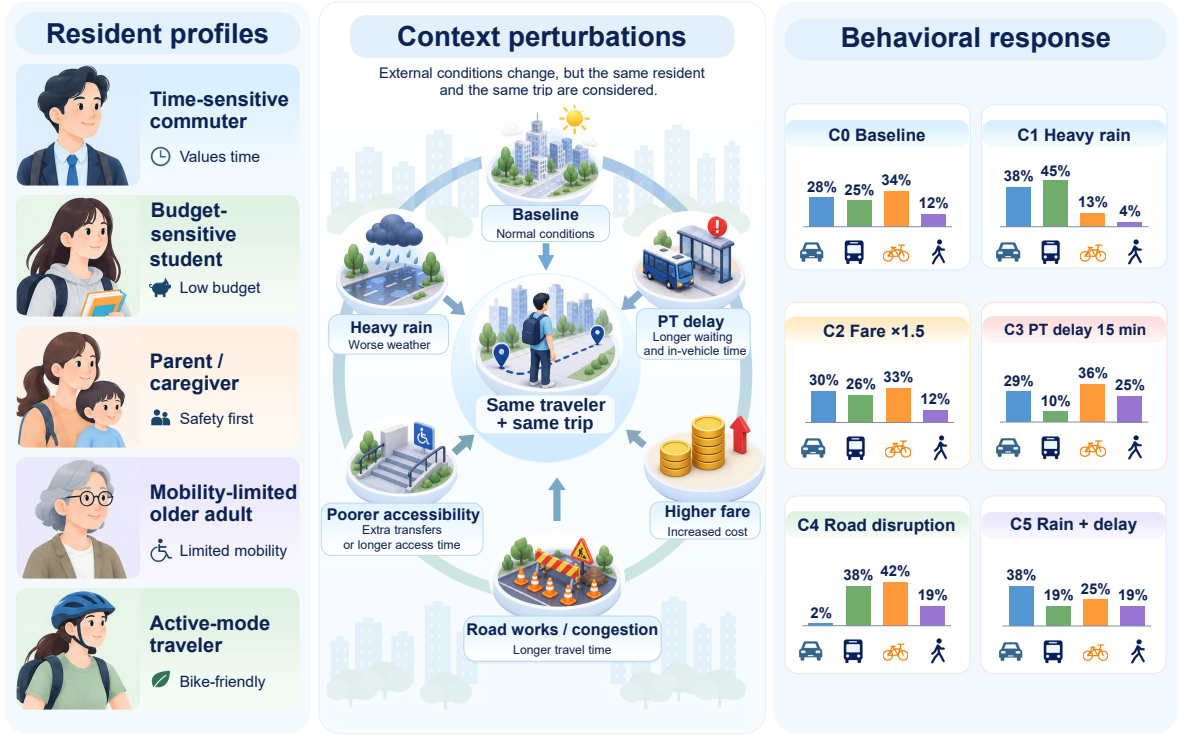
The distinction matters because predictive agreement is not behavioral agreement. In travel choice modeling, models with strong predictive performance can differ in elasticities and other behavioral indicators (Wang et al., 2020; Martín-Baos et al., 2023). LLM-based travel studies likewise distinguish plausible responses from empirically aligned behavior (Yan et al., 2026; Liu et al., 2026c; Xu and Wang, 2026). Distillation introduces another layer: even a Student that resembles its Teacher at individual states may differ in how its predictions change between states. Conversely, a less faithful Student may accidentally agree more closely with human responses. Neither model compression nor improved prediction alone resolves this issue.

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We distinguish *static fidelity*, agreement with the Teacher at individual states, from *response fidelity*, agreement in finite changes of elicited choice distributions for the same traveler and trip. These contrasts are defined for specified perturbations; they are not, by themselves, population causal effects or elasticities. Figure 1 illustrates the traveler–trip pairing and its connection to aggregate demand. Exact agreement at every endpoint would imply exact response agreement. In finite data and finite-capacity models, however, average endpoint scores do not directly evaluate the contrasts that a scenario analysis queries.



**Figure 1:** Traveler–trip pairing and illustrative demand responses. Each pair changes travel conditions while holding the traveler and trip fixed. Profiles are illustrative; bars are rounded, pre-routing mode assignments for 10,000 synthetic Singapore travelers, not survey observations. They use the supply-adapted model defined in Section 3.2. Poorer accessibility is evaluated separately. Scenario definitions and exact shares are in Supplementary Table S29.

Response fidelity is a criterion for faithful compression, not a guarantee of behavioral validity. A Student can inherit a Teacher’s discrepancy from people. Independently, a predicted change can be altered by the rule used to assign modes, by route search and by simulation execution. The central question of this study is therefore whether model performance carries across three evaluation levels: *Teacher-response fidelity*, *agreement with stated human responses*, and *preservation through network execution*. These levels are complementary tests, not interchangeable definitions of success.

The study makes three contributions. First, it operationalizes response-centric evaluation with a persona-disjoint benchmark and matched neural objectives, alongside a structured logit Student and separate timing predictors. This design distinguishes the effect of explicit response weighting from that of model specification. Second, independent stated-choice samples in Singapore and Shanghai test whether Teacher-response fidelity orders models in the same way as agreement with people. A later, same-task LLM comparison provides a descriptive reference, separately from the archived training targets. Third, a controlled Helsinki MATSim experiment traces each response from predicted probabilities to PT boarding, separating assignment and execution effects from changes in the behavioral model. These are linked evaluations of model behavior, not a claim of uniform geographic transfer across the three settings.

The strongest finding is a change in what counts as the best model. Response-aware supervision improves Teacher-response error modestly; a linear choice specification improves it more. Yet the Student closest to the Teacher reverses a stated accessibility response, and an assignment rule that improves routability does not necessarily improve response

preservation. The contribution is thus a validation procedure and empirical diagnosis for LLM-derived traveler models, with learning objectives and modular Students as testable implementations. It identifies which comparison must be passed before an imitation score can support a transport-simulation claim.

## 2. Related work

### 2.1. LLM traveler models and empirical alignment

LLMs have been used to generate activities and travel in urban simulation (Liu et al., 2026a,b), model day-to-day route choice (Wang et al., 2025), and predict modes using retrieved examples or incremental learning (Xu and Jiao, 2027; Chen et al., 2026). Conceptual work has set out their possible roles within transport systems and motivated hybrid designs (Liu et al., 2025; Nie et al., 2025). These applications establish the relevance of LLM-generated decisions; they do not make fidelity to an LLM equivalent to fidelity to travelers.

Empirical alignment is a separate research problem. Studies of synthetic survey respondents document both reproducible aggregate patterns and sensitivity to prompts and model versions (Argyle et al., 2023; Aher et al., 2023; Bisbee et al., 2024). For travel choice, Yan et al. (2026) examine the direction and magnitude of time trade-offs, while Liu et al. (2026c) learn persona-based representations using empirical choices. In this journal, Xu and Wang (2026) compare zero-shot and few-shot LLM predictions of travel satisfaction and study improvements from task-specific examples. Learning and adjustment behavior have also been addressed explicitly (Liu et al., 2026d). These studies motivate empirical checks rather than treating the Teacher as a behavioral ground truth.

The distinction in the present study is the object being audited: a numerical Student fitted to archived LLM judgments. We evaluate whether its finite responses match those judgments and whether this ranking transfers to independently collected stated responses. The human comparison is an external evaluation, not a calibration procedure. The contemporary LLM subset is a later reference model and does not, without matched backend provenance, identify how much discrepancy originated during compression.

### 2.2. Predictive and behavioral criteria for choice models

Random-utility models connect choice probabilities to traveler characteristics and alternative attributes (McFadden, 1974; Ben-Akiva and Lerman, 1985). Generalized and mixed logit formulations represent richer heterogeneity (Walker and Ben-Akiva, 2002; Hensher and Greene, 2003; Train, 2009), while representation learning can retain a specified utility core alongside flexible components (Siffringer et al., 2020). Comparisons of these families show that predictive performance, market shares and behavioral indicators can favor different models (Wang et al., 2020; Martín-Baos et al., 2023).

Distillation changes the estimation target from observed choices to another model's elicited distributions. A structured utility model may preserve such responses through regularization and restricted functional form, without demonstrating that the Teacher or travelers follow that form. We therefore compare logit and neural specifications on the same information, then evaluate their human-response discrepancies separately. Independent timing predictors test whether choice and departure adjustment benefit from different specifications rather than assuming that a shared representation is preferable for both.

### 2.3. Response supervision and simulation surrogates

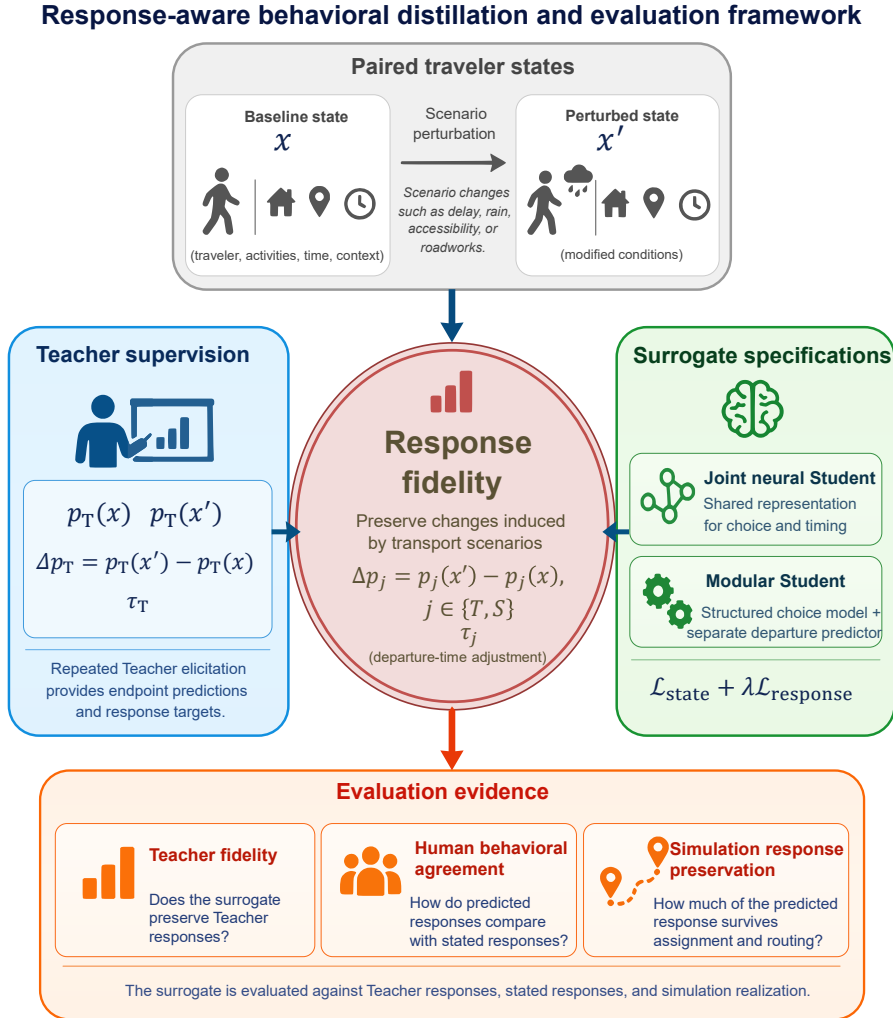
Knowledge distillation transfers predictions or representations to a smaller model (Hinton et al., 2015; Gou et al., 2021; Sanh et al., 2019). Relational distillation preserves relationships between examples (Park et al., 2019); Sobolev training uses output and derivative information (Czarnecki et al., 2017). Response-aware fitting is therefore not a new general principle. Here the relation is a finite, travel-specific difference between elicited distributions. It is calculated from endpoint targets already available to every controlled Student, so the experiment concerns an optimization preference for scenario contrasts rather than access to additional Teacher information.

Surrogate modeling also has an established transport role. Natterer et al. (2025) approximate network outcomes of a MATSim model to accelerate scenario exploration. Our Student instead replaces the traveler decision function while retaining network construction and execution. This distinction makes assignment and routing part of the evaluation rather than outputs that the surrogate directly predicts. A concurrent preprint, CALM, combines calibrated choice, optional LLM planning, network feedback and replayable evaluation (Cheng, 2026). Its system-level integration is complementary to our narrower question: whether finite responses survive compression and whether the resulting

model ordering persists across Teacher, stated-choice and execution comparisons. Neither the use of an LLM in a simulator nor the presence of intervention tests alone constitutes the contribution claimed here.

### 3. Response-centric evaluation and distillation

Behavioral distillation is treated here as the estimation of a surrogate that reproduces both the Teacher's predictions at individual traveler states and its reaction to specified changes in travel conditions. Every candidate model produces a mode-choice distribution and a departure-time adjustment, which allows joint and modular specifications to be compared on the same behavioral outcomes. How these outcomes are realized in simulation is examined separately (Figure 2).



**Figure 2:** Response-aware behavioral distillation and evaluation. Paired states define changes in mode-choice probabilities, while the Teacher also supplies departure-adjustment targets. Joint neural and modular Students predict choice and timing. The schematic objective combines endpoint fitting with response supervision; the four controlled objectives are specified in Section 3.3, and MNL-S is fitted to endpoint targets. SA-Student additionally uses supply and between-traveler supervision outside that matched comparison. Evaluation distinguishes Teacher fidelity, agreement with stated responses and preservation of predicted responses through assignment, routing and MATSim execution. Arrows link these components conceptually; human responses are used only for evaluation, and simulated experience does not feed back into the behavioral predictions.

### 3.1. Traveler states and response targets

A state  $\mathbf{x}$  describes a traveler, a trip and its context, together with level-of-service attributes for four alternatives: private car, public transport (PT), bicycle and walking. The attributes cover travel time, monetary cost and weather exposure, and for PT also access, waiting, in-vehicle and transfer conditions. Variable definitions and encoding rules are given in Supplementary Section S1 and Table S6.

Let  $\mathbf{p}_j(\mathbf{x})$  denote the elicited or predicted choice distribution and  $\tau_j(\mathbf{x})$  the departure adjustment in minutes, where  $j \in \{T, S\}$  indexes the Teacher and the Student. The term probability refers to the normalized model output, not an established human choice frequency. For a pair of states that share the same traveler and trip,

$$\Delta \mathbf{p}_j(\mathbf{x}, \mathbf{x}') = \mathbf{p}_j(\mathbf{x}') - \mathbf{p}_j(\mathbf{x}), \quad j \in \{T, S\}. \quad (1)$$

These differences are the response targets. The perturbation from  $\mathbf{x}$  to  $\mathbf{x}'$  is a specified change to the state, such as additional PT delay or heavy rain. The resulting finite contrasts describe how a model reacts to that change. They are not, on their own, estimates of elasticities or of causal effects in a traveler population.

We used DeepSeek V4 Pro, supplied directly through the official DeepSeek API (<https://api.deepseek.com>), for Teacher elicitation throughout the study, with the request model `deepseek-v4-pro`. Valid repeated numerical probability vectors are averaged within state; repetition measures elicitation variability rather than additional travelers. The service and model attribution is based on the authors' collection records and confirmation; retained model fields in the later campaigns also report `deepseek-v4-pro`. The provider's current documentation identifies this service as DeepSeek-V4-Pro-0813 and confirms continued Pro API service after 14 September 2026 (DeepSeek, 2026b,a). This service-level attribution does not establish an immutable backend fingerprint for every archived response. We retain the distinction between frozen training targets and later same-task elicitations; Supplementary Table S38 gives their provenance.

### 3.2. Joint and modular behavioral specifications

The joint neural Student learns choice and timing from shared representations. A global encoder embeds traveler and trip attributes, an alternative encoder embeds mode-specific attributes, and a shared scorer maps both to a utility  $u_m$ . Probabilities are normalized over the alternatives available to the traveler,  $\mathcal{A}(\mathbf{x})$ :

$$p_{S,m}(\mathbf{x}) = \frac{\exp(u_m)}{\sum_{k \in \mathcal{A}(\mathbf{x})} \exp(u_k)}, \quad m \in \mathcal{A}(\mathbf{x}). \quad (2)$$

Unavailable modes receive zero probability. Departure adjustment is predicted from the global representation and the mean representation of the available alternatives, with  $\tau_S = 60 \tanh(z)$  limiting the adjustment to  $\pm 60$  minutes. Network dimensions are reported in Supplementary Section S2.

The supply-matched multinomial logit Student (MNL-S) replaces the neural scorer with four linear utilities on the same structured information and is fitted to the soft Teacher targets. Choice and timing are estimated separately in this specification. Three timing predictors are paired with the same MNL-S choice probabilities: ridge regression, a bounded linear predictor  $60 \tanh(\mathbf{w}^\top \mathbf{x} + b)$ , and an independent neural predictor that shares the joint model's encoder structure but receives only departure supervision. Holding the choice model fixed isolates the contribution of the timing specification.

A third model, the *supply-adapted Student* (SA-Student), extends a pretrained joint model with additional PT-supply attributes and with supervision on contrasts between travelers, between contexts and attributes, and between accessibility levels. The adapted model was selected by minimizing KL divergence on the accessibility validation states, subject to constraints on its earlier performance (Supplementary Table S25), and it is held fixed in the stated-choice and simulation experiments. Because its initialization, objective, auxiliary supervision and treatment of mode availability all differ from those of the controlled models, SA-Student is reported as an additional model rather than as a test of the response loss. The roles of the three model families are summarized in Supplementary Table S36, and the adaptation is specified in Section S4.

### 3.3. Response-aware learning objectives

Four objectives are compared within the joint architecture. With  $H_1$  denoting the Huber loss with a one-minute transition, the objective at a single state is

$$\ell_{\text{soft}}(\mathbf{x}) = D_{\text{KL}}(\mathbf{p}_T \| \mathbf{p}_S) + H_1(\tau_S - \tau_T). \quad (3)$$

*Soft KL* applies this loss to both states of every training pair. *CE+KL* adds an equally weighted cross-entropy term for the Teacher's most likely mode, a hard label derived from its probability vector. The two response-aware objectives instead add a penalty on probability changes to soft KL and omit the hard label.

Within a training batch  $\mathcal{B}$ , let  $a_{im}$  indicate that mode  $m$  is available in at least one state of pair  $i$ . The *signed-response* loss is

$$\ell_{\text{signed}} = \frac{\sum_{i \in \mathcal{B}, m} a_{im} |\Delta p_{S,im} - \Delta p_{T,im}|}{\max(1, \sum_{i \in \mathcal{B}, m} a_{im})}. \quad (4)$$

The *direction+magnitude* objective penalizes the sign and the size of the response separately:

$$\ell_{\text{dir}} = \frac{\sum_{i \in \mathcal{B}, m} a_{im} b_{im} [-\text{sgn}(\Delta p_{T,im}) \Delta p_{S,im}]_+}{\max(1, \sum_{i \in \mathcal{B}, m} a_{im} b_{im})}, \quad (5)$$

$$\ell_{\text{mag}} = \frac{\sum_{i \in \mathcal{B}, m} a_{im} \left| |\Delta p_{T,im}| - |\Delta p_{S,im}| \right|}{\max(1, \sum_{i \in \mathcal{B}, m} a_{im})}, \quad (6)$$

where  $b_{im} = \mathbf{1}(|\Delta p_{T,im}| > 10^{-6})$  and  $[u]_+ = \max(u, 0)$ . Both terms carry unit weight. Their denominators differ because the direction term ignores negligible Teacher responses. The auxiliary supervision of SA-Student plays no part in this comparison.

Paired supervision changes the geometry of the fitting objective: it prioritizes differences between states whose endpoint targets are shared by all controlled models. Thus any improvement is attributable to response weighting under the matched protocol, not to additional Teacher information. Writing  $\mathbf{e}(\mathbf{x}) = \mathbf{p}_S(\mathbf{x}) - \mathbf{p}_T(\mathbf{x})$  for the endpoint error,

$$\|\Delta \mathbf{p}_S - \Delta \mathbf{p}_T\|_1 \leq \|\mathbf{e}(\mathbf{x}')\|_1 + \|\mathbf{e}(\mathbf{x})\|_1, \quad (7)$$

so a Student that matched the Teacher exactly at every state would also reproduce every response. The experiment tests response weighting under finite coverage and capacity. These penalties are optimization targets, not directional constraints: equally normalized direction and magnitude terms can have cancelling gradients on an interval of wrong-sign responses, and CE+KL changes the idealized single-state target. Supplementary Section S2.3 derives both properties. The direct signed-response objective supplies a comparison that does not rely on splitting sign and magnitude.

### 3.4. From predicted behavior to simulated demand

Behavioral availability and route feasibility restrict choices in different ways. Availability defines the choice set in Eq. 2, whereas route feasibility depends on whether the route search finds a connection for a given trip and departure time. The PT search considers direct and one-transfer connections, so an unsuccessful search does not prove that no connection exists (Supplementary Section S1).

Three rules convert predicted probabilities into assigned modes. Deterministic assignment selects the most probable mode, stochastic assignment samples from the predicted distribution, and feasibility-constrained assignment samples after removing probability from alternatives for which no route was found and renormalizing the remainder. Without this constraint, a failed route search for car, bicycle or PT triggers a direct fallback to walking, rather than selection of the next-highest-probability mode. The walking route is then constructed on the walking network.

The feasibility correction has an explicit distributional cost. Let  $F$  be the feasible set and  $Z = \sum_{m \in F} p_m > 0$  the retained probability mass. Conditional renormalization gives  $q_m = p_m / Z$  for  $m \in F$  and zero otherwise, satisfying

$$\|\mathbf{q} - \mathbf{p}\|_1 = 2(1 - Z) = \min_{\mathbf{r} \in \Delta(F)} \|\mathbf{r} - \mathbf{p}\|_1, \quad (8)$$

where  $\Delta(F)$  is the simplex supported on  $F$ . Thus a feasible assignment distribution cannot preserve positive mass on an excluded mode. Minimizing this endpoint displacement does not minimize response displacement: with  $\mathbf{d} = \mathbf{q} - \mathbf{p}$ , the change induced in a response is  $\mathbf{d}(\mathbf{x}') - \mathbf{d}(\mathbf{x})$ . The identity is a property of the assignment rule, not a new learning result; a proof and counterexample are given in Supplementary Section S5.4.

For each traveler, supply attributes are computed at the requested departure time. The fixed model predicts a departure adjustment, which is added before checking route feasibility. The feasibility mask requires behavioral availability and a route returned in the requested mode at this adjusted time. Feasibility-constrained assignment applies

this mask and renormalizes once; final routing uses the same adjusted time and routing procedure, without another behavioral prediction. Return trips are routed separately at their scheduled return time, with the same fallback rule. If the retained probability mass is zero, the implementation stops the run rather than dropping that traveler or inventing an alternative. All 140 completed runs retained the full population; none encountered zero feasible mass. Supplementary Section S7 reports the route-ledger checks and implementation details.

PT use is then traced through five stages: predicted probabilities, feasibility-adjusted probabilities, assigned modes, routed modes and PT boarding in the simulation. Every stage uses the initial population as its denominator. For a contrast  $c$ , the total response gap can be separated as

$$G_c = R_c^{\text{sim}} - R_c^{\text{pred}} = \underbrace{R_c^{\text{adj}} - R_c^{\text{pred}}}_{\text{feasibility adjustment}} + \underbrace{R_c^{\text{sim}} - R_c^{\text{adj}}}_{\text{downstream execution}}, \quad (9)$$

where  $R_c^{\text{adj}}$  is the response after the specified probability correction. Without that correction,  $R_c^{\text{adj}} = R_c^{\text{pred}}$ . This separates the effect of changing the sampling distribution from subsequent assignment and execution. Simulated PT use means boarding a PT vehicle before reaching the outbound destination, which is not the same as completing the journey. Because the behavioral predictions are held fixed during simulation, the gap reflects assignment, routing and simulation execution with fixed behavioral predictions.

## 4. Experimental design

Each of the three comparisons in Section 1 draws on a different kind of evidence, summarized in Table 1. The controlled benchmark isolates the effect of the learning objective and compares model specifications against Teacher targets. The stated-choice experiments assess agreement with people on data that play no part in fitting or model selection, and the simulation experiment follows predicted responses through assignment and routing.

**Table 1**

Evaluation questions, evidence units and comparison scope.

| Question                        | Evidence units                                        | Comparison and scope                                                                                                                                         |
|---------------------------------|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RQ1: Teacher-response fidelity  | Six held-out synthetic personas; three training seeds | Four neural objectives trained under matched conditions isolate the effect of the loss. MNL-S with separate timing predictors compares model specifications. |
| RQ2: Stated-response agreement  | 332 Singapore and 321 Shanghai respondents            | Within-person choice contrasts compared with the responses of the fitted models.                                                                             |
| RQ2: Teacher comparison         | 24 respondents per city                               | Contemporary Teacher and Student responses on the same tasks, decomposed into Teacher and Student discrepancies.                                             |
| RQ3: Preservation in simulation | 1,000 fixed Helsinki travelers                        | Each model family combined with three assignment rules under a perceived PT delay, simulated in MATSim.                                                      |

### 4.1. Controlled Teacher-response benchmark

A persona is a generated traveler profile, and a state combines a persona with a trip, a context, alternative attributes and Teacher targets. The training, validation and test sets contain 1,839, 365 and 437 states drawn from disjoint groups of 28, six and six personas. The test states form 398 response pairs and 94 interaction contrasts. Single-context changes, PT-supply profiles, joint perturbations and context–attribute contrasts all follow the same persona partition (Supplementary Table S7). The prepared manifest also excludes the fare–congestion joint combination from training and validation (112 and 24 candidate states, respectively). This provides a limited compositional holdout within the joint source; the reported aggregate comparison does not establish generalization across all unseen intervention families or intensity ranges.

Within each of three training seeds, the four neural objectives share the architecture, inputs, availability rules, training pairs, initialization and optimization budget, so that the objective is the only difference between them. Models for the probability-response comparison are selected by macro-source KL, a validation KL divergence summarized at the level of the four supervision sources. MNL-S receives the same endpoint information and exposure weights,

is regularized separately and is selected by the same criterion. Optimization settings are reported in Supplementary Section S2.

Departure prediction is evaluated in a separate comparison selected by validation departure MAE. This criterion chooses the bounded-linear and neural timing predictors, the ridge penalty, and independently trained versions of the four joint objectives. The neural and bounded predictors share data, information and update budget, which separates the effect of the timing specification from that of the selection rule.

Response fidelity is measured as the absolute difference between Teacher and Student probability changes, averaged first over the modes available in a pair and then over pairs. Macro-source KL, departure MAE against the Teacher and interaction error complement this measure. Interaction error compares the response to a combined perturbation with the responses to its components. Formal definitions, including the feasibility violation rate, are given in Supplementary Section S2.1.

Two sources of uncertainty are reported. Training-seed standard deviations describe optimization variability, whereas paired intervals clustered by persona describe variation across the six held-out personas and are conditional on them. Consistency within the benchmark is further examined through persona-level results, paired seed differences, removal of each test persona from evaluation without retraining, and reaggregation of the repeated Teacher outputs that are available.

## 4.2. Stated-choice experiments

The Singapore sample comprises 332 eligible residents who made 3,320 choices. Its ten tasks form five within-person pairs on PT accessibility, rain, fares, PT delay and roadworks. The Shanghai cohort evaluated by the models comprises 321 respondents with 3,208 explicit four-mode choices over ten tasks. Two further answers indicate no suitable mode and are not scored. Twelve eligible respondents whose answers fall outside the model's input mapping are retained in a human-only sensitivity analysis. Responses were collected from 20 August to 5 September 2026 in Singapore and from 12 to 18 September 2026 in Shanghai, and both questionnaires recorded participant consent. We distributed electronic questionnaires through WeChat, WhatsApp and related personal contacts to research-group members' families, friends, classmates and colleagues, and through onward sharing within those networks. Singapore used Google Forms and Shanghai used Wenjuanxing. The target of inference is response agreement within these convenience and snowball samples, not citywide mode shares. Singapore eligibility required residence in Singapore; Shanghai eligibility required travel in Shanghai during the preceding 30 days. Participants completed ten stated-choice tasks online. Sample flow and questionnaire access are reported in Supplementary Section S1.

*Ethics and consent statement.* The stated-choice questionnaires were administered as anonymous, minimal-risk surveys. No direct personal identifiers were collected. Before beginning the questionnaire, all participants were shown an electronic information and consent statement describing the study purpose, voluntary participation, intended research use of the responses, and confidentiality protections; only respondents who provided consent proceeded to the survey. Formal institutional ethics approval was not obtained. The study procedures were designed to protect participant privacy and to limit data collection to information necessary for the stated research purpose.

The two surveys specify their tasks in different ways. Singapore shows respondents numerical tables of time and cost. Shanghai describes most conditions qualitatively, with explicit values only for a 6-yuan PT fare, a 30-yuan parking fee and one transfer. Model inputs therefore combine attributes displayed in the questionnaire, characteristics reported by respondents and reference values for information that was not collected (Supplementary Table S37). The reference values do not depend on respondents' answers. Twelve alternative specifications for Singapore and eight for Shanghai vary selected missing attributes and reference times (Supplementary Table S4).

The questionnaire's car option is represented by the model's private-car alternative. Private car requires ownership and a licence, and bicycle requires access. All explicit choices are scored, including eight Singapore and two Shanghai choices of a mode that the model treats as unavailable, so that the evaluation sample is not reduced to choices the model can represent.

For respondent  $i$ , contrast  $c$ , task weights  $w_{ct}$  and target mode group  $M_c$ , the stated and predicted responses and the response bias are

$$r_{ic}^H = \sum_t w_{ct} \mathbf{1}(y_{it} \in M_c), \quad (10)$$

$$r_{ic}^j = \sum_t w_{ct} \sum_{m \in M_c} p_{itm}^j, \quad j \in \{T, S\}, \quad (11)$$

$$B_c^j = \text{mean}_i(r_{ic}^j - r_{ic}^H). \quad (12)$$

Simple pairs use weights  $-1$  and  $+1$ , and the rain–delay interaction combines four tasks. Intervals come from a respondent bootstrap that resamples each person’s full set of tasks, with shared draws for paired model comparisons. Nominal 95% intervals are reported together with Bonferroni-adjusted intervals over the five Singapore or eight Shanghai contrasts, computed separately for each model and city. These intervals are conditional on the fitted models, and an interval that includes zero does not establish equivalence. Choice accuracy, mode shares, Brier score and negative log-likelihood (NLL) complement the response measures. Because a mode with zero predicted probability makes NLL infinite, NLL is computed with explicit probability truncation (Supplementary Section S3).

A further Shanghai analysis asks whether inputs derived from a transport network improve agreement because they are matched to individual respondents. Origin–destination pairs of approximately 6 km are assigned using OSM roads and a synthetic metro timetable, and the assignment is then repeated and complete supply profiles are exchanged between respondents. The constructed trips test sensitivity to numerical inputs and are not observations of respondents’ journeys.

### 4.3. Contemporary API reference on the survey tasks

A contemporary API model is queried on the stated-choice tasks for 24 respondents per city, stratified by age and car ownership. This provides a same-input comparison with the frozen Students; it is distinct from the archived Teacher targets used for training. Each of the ten tasks per respondent receives three successful Teacher elicitations. The Teacher receives the same numerical and categorical attributes and the same availability rule as the Students, but not the human answers. Metadata that the Students do not use are set to neutral values, and the prompt explains PT feasibility and its time components. Service settings and repeat variability are reported in Supplementary Section S3.2.

For this subsection,  $T$  denotes the contemporary API reference,  $T_{\text{now}}$ , rather than a verified reconstruction of the training backend. On the same respondents, the signed response discrepancy decomposes as

$$\bar{r}_c^S - \bar{r}_c^H = (\bar{r}_c^T - \bar{r}_c^H) + (\bar{r}_c^S - \bar{r}_c^T). \quad (13)$$

The first term compares the current API reference with stated behavior; the second compares the Student with that reference. This is an exact descriptive identity, not an identification of compression error. Both campaigns used the official DeepSeek V4 Pro service, but without a common immutable backend record the second term can also include differences between acquisition periods and elicitation protocols (Bisbee et al., 2024). All terms use common respondent resamples, with elicitation variability reported separately.

### 4.4. Simulation and computational experiments

The Helsinki experiment compares a baseline and a perceived-delay condition for a fixed population of 1,000 travelers, using OSM roads and HSL timetables. The delay condition adds 15 minutes to the PT delay perceived by the behavioral model while leaving the physical timetable unchanged. SA-Student, the soft-KL and direction+magnitude neural Students, and MNL-S with independent neural timing are each combined with the three assignment rules of Section 3.4. Common random numbers are used across paired conditions and assignment rules, and training and assignment variability are summarized separately (Supplementary Table S26).

At each stage, the response  $R$  is the change in PT probability or PT-use share from baseline to delay over the same 1,000 people. A single MATSim iteration with fixed behavioral predictions isolates the gap  $G$  introduced by assignment, routing and execution. Holding the timetable fixed is the control that separates execution effects from changes in physical supply. The estimand is response transmission through this pipeline, not adaptation to a physical service disruption or equilibrium demand. Additional comparisons route trips at desired and at adjusted departure times. Walking and cycling use network-derived link speeds, and a Shanghai experiment examines alternative active-mode speed assumptions while keeping people, assigned modes, departure times and plans fixed.

A Singapore experiment records the time needed to construct simulation scenarios for nested populations of up to 50,000 agents, separating behavioral inference from feature and route preparation and measuring construction with and without reuse of unchanged supply calculations. Teacher latency is measured on successful requests only, and population-scale Teacher times are projections rather than measured builds (Supplementary Sections S5 and S6).

**Table 2**

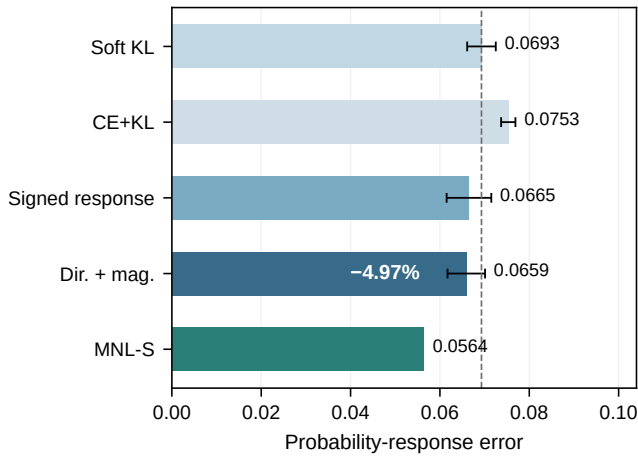
Probability-response fidelity on the held-out Teacher benchmark.

| Model / objective   | Macro-source KL ↓   | Response error ↓    | Interaction error ↓ |
|---------------------|---------------------|---------------------|---------------------|
| Soft KL             | $0.0915 \pm 0.0187$ | $0.0693 \pm 0.0032$ | $0.0457 \pm 0.0038$ |
| CE+KL               | $0.1046 \pm 0.0170$ | $0.0753 \pm 0.0016$ | $0.0518 \pm 0.0038$ |
| Signed response     | $0.0871 \pm 0.0138$ | $0.0665 \pm 0.0050$ | $0.0465 \pm 0.0036$ |
| Direction+magnitude | $0.0861 \pm 0.0145$ | $0.0659 \pm 0.0042$ | $0.0464 \pm 0.0039$ |
| MNL-S               | 0.0731              | 0.0564              | 0.0386              |

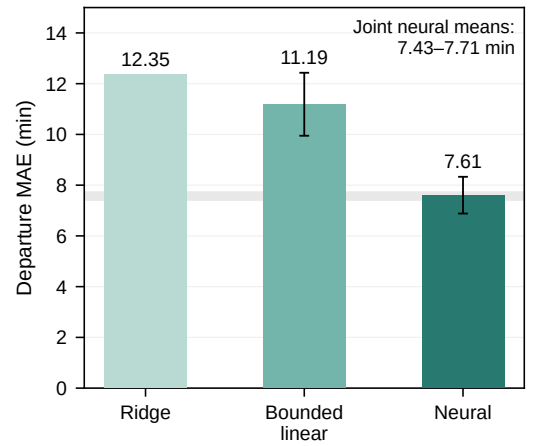
Neural entries are means  $\pm$  SD over three training seeds. The test set contains 437 states, 398 response pairs and 94 interaction contrasts from six held-out synthetic personas. All models, including the MNL-S regularization, are selected by validation macro-source KL. MNL-S uses the same inputs and training states as the neural Students.

**(a) Response supervision**

Models selected by validation KL

**(b) Modular departure prediction**

Fixed MNL-S choice probabilities



**Figure 3:** Response learning and modular departure prediction. (a) Probability-response errors for models selected by validation KL. The dashed line marks soft KL. (b) Departure MAE of three timing predictors combined with the same MNL-S choice model, whose probabilities are identical across the three. The shaded band spans the means of the four joint neural models selected for timing (7.43–7.71 min). Error bars show training-seed SD where applicable. Departure targets are the Teacher’s.

**Table 3**

Departure prediction under a common selection criterion (validation MAE).

| Choice model                | Departure predictor    | Total parameters | MAE (min) ↓      |
|-----------------------------|------------------------|------------------|------------------|
| Joint / soft KL             | Shared neural encoders | 24,562           | $7.61 \pm 0.75$  |
| Joint / CE+KL               | Shared neural encoders | 24,562           | $7.43 \pm 0.91$  |
| Joint / signed response     | Shared neural encoders | 24,562           | $7.71 \pm 0.66$  |
| Joint / direction+magnitude | Shared neural encoders | 24,562           | $7.55 \pm 0.80$  |
| MNL-S                       | Ridge regression       | 555              | 12.35            |
| MNL-S                       | Bounded linear         | 555              | $11.19 \pm 1.24$ |
| MNL-S                       | Independent neural     | 18,733           | $7.61 \pm 0.72$  |

Errors are measured against Teacher departure targets on the 437 test states. Neural and bounded-linear predictors report means  $\pm$  SD over three training seeds, and the ridge fit is deterministic. MNL-S choice probabilities are identical across the three timing predictors. The joint models in this table are trained separately and selected by validation departure MAE, whereas those in Table 2 are selected by validation KL.

## 5. Results

### 5.1. Response fidelity and behavioral specification

Explicit response supervision improves finite-contrast fidelity under the matched benchmark protocol. Direction+magnitude lowers mean response error from 0.06930 under soft KL to 0.06586, a reduction of 4.97% (Table 2; Figure 3a). The paired difference is  $-0.00345$ , with a persona-clustered 95% interval of  $[-0.00504, -0.00191]$ . Signed response reaches 0.06653. The difference between the two paired objectives is small ( $-0.00067$ , interval  $[-0.00199, 0.00072]$ ), which supports explicit response supervision but does not establish an additional benefit from separating sign and size.

The gain is distributed across the tested seeds and personas. Across the three training seeds, direction+magnitude differs from soft KL by  $-0.00499$ ,  $-0.00441$  and  $-0.00094$ , although the interval for the third seed includes zero. Averaged over seeds, the error is lower for each of the six test personas, and removing any one persona from evaluation leaves differences between  $-0.00408$  and  $-0.00310$  (Supplementary Tables S11 and S13). The advantage of both paired objectives also persists when the targets are recomputed from the available repeated Teacher outputs, an analysis that excludes the mechanism states for which repeats were not recovered.

The benefit is specific to single contrasts. For combined perturbations, the paired objectives reach interaction errors of 0.0465 and 0.0464, slightly above the 0.0457 of soft KL, so supervising individual responses did not improve interaction error in this benchmark. CE+KL has the largest response error, 0.07528, even though it fits an additional target. Its hard label pulls predictions toward the Teacher's leading mode, and in the fitted networks this enlarges the probability changes on pairs whose leading mode does not change (Supplementary Section S2.3). Fitting the most likely mode appears to distort the graded responses on which scenario analysis depends.

A change of specification matters more than any change of objective. MNL-S, fitted to the same endpoints with linear utilities, attains the lowest errors on all three measures: 0.05644 for response error, 0.07310 for macro-source KL and 0.03865 for interaction error. Its response error is 14.30% below that of direction+magnitude, with a paired interval of  $[-0.01246, -0.00722]$ . One plausible reading is that structured utilities restrict how a change in an attribute can propagate to probabilities. With only 28 training personas, this restriction appears to generalize better than the additional freedom of the neural Students, which can fit individual states without reproducing the contrasts between them.

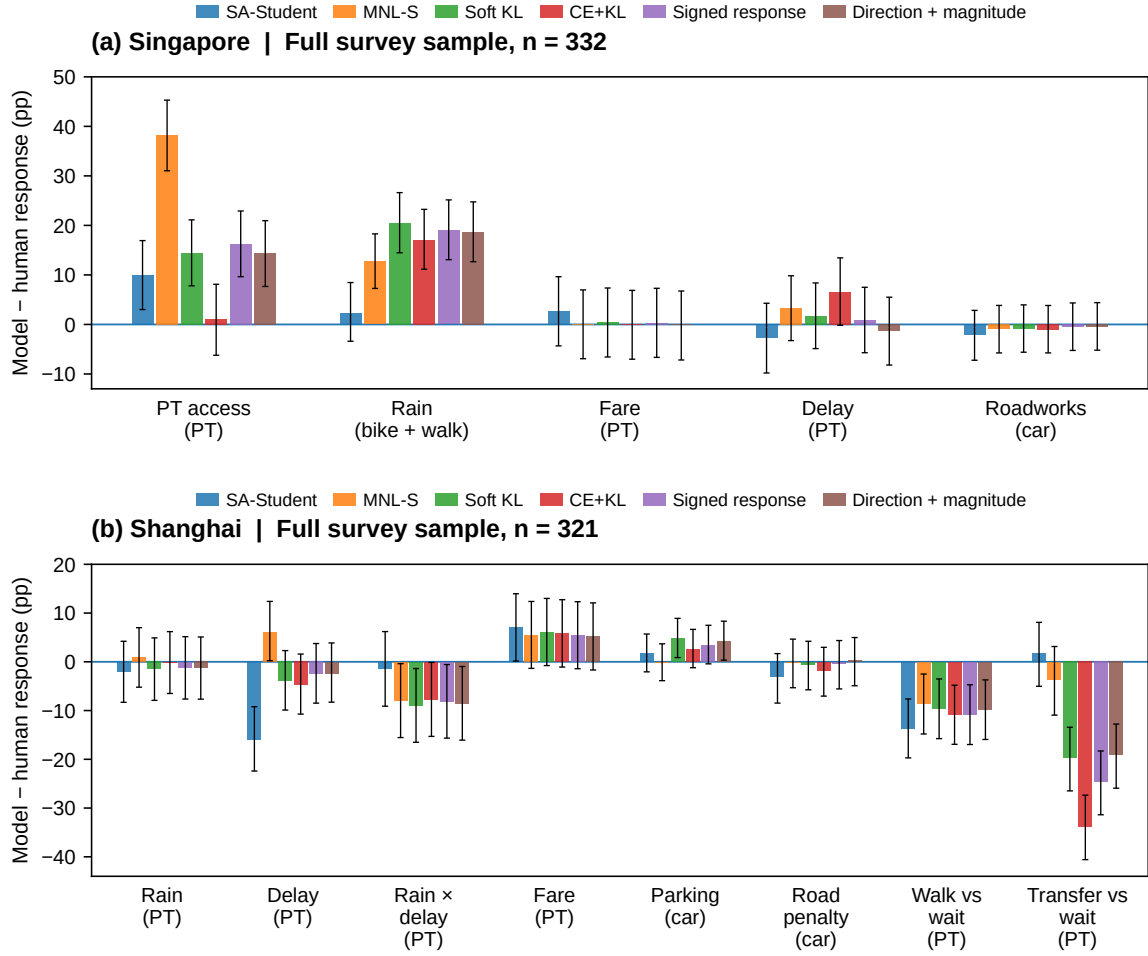
The linear specification is less suited to timing. With the MNL-S choice probabilities held fixed, ridge regression, the bounded linear predictor and the independent neural predictor give departure MAEs of 12.35, 11.19 and 7.61 minutes (Table 3; Figure 3b). The neural predictor brings the modular model within the 7.43–7.71-minute range of the joint models selected for timing, with 18,733 rather than 24,562 parameters. Choice and timing therefore favor different functional forms, and estimating them separately retains the response advantage of linear utilities with comparable departure errors in this benchmark. All timing errors are measured against Teacher targets, not observed departure behavior.

### 5.2. Agreement with stated responses

How well a Student agrees with people depends on the intervention. In Singapore, SA-Student predicts a PT-delay response of  $-14.86$  percentage points, compared with a stated change of  $-12.05$  points, a bias of  $-2.81$  points (nominal 95% interval  $[-8.18, 2.75]$ ). Rain lowers its bicycle-plus-walk probability by 19.70 points, close to the 21.99-point fall in stated choice. Poorer PT accessibility is a different case. The Student predicts a change of  $-14.42$  points against a stated  $-24.40$ , and the resulting  $+9.98$ -point bias remains positive after adjustment for multiple contrasts ( $[3.01, 16.93]$ ).

Shanghai reverses the pattern for delay. Under the reference specification, SA-Student predicts a PT-delay response of  $-30.57$  points, roughly twice the stated  $-14.64$ . The bias of  $-15.93$  points excludes zero after adjustment over eight contrasts ( $[-22.41, -9.21]$ ). The Student also predicts responses of the wrong sign for the fare and walking-versus-waiting contrasts (Figure 4). These discrepancies persist across the eight alternative input specifications, where the delay bias ranges from  $-16.53$  to  $-10.25$  points and both sign reversals remain. These specifications retain the 15-minute encoding of the qualitative delay condition; they do not test sensitivity to its intensity.

The ranking of objectives on the Teacher benchmark does not carry over to people. In Singapore, direction+magnitude shifts the delay bias from  $+1.78$  points under soft KL to  $-1.34$ , yet both objectives substantially understate the rain response. MNL-S offers the clearest example. It approximates the Singapore fare response and the Shanghai parking and road responses, but under poorer Singapore accessibility it predicts a PT change of  $+13.85$  points, whereas the stated PT share falls by 24.40 points. The accessibility task raises PT travel time from 32 to 48



**Figure 4:** Response differences in the full samples, defined as the change in model probability minus the change in stated-choice share. Whiskers show Bonferroni-adjusted respondent-bootstrap intervals within each city (99% for Singapore and 99.375% for Shanghai). Neural means average three fitted seeds. Shanghai results use the reference specification of Section 4.2. The interaction and walking-versus-waiting contrasts include 320 respondents.

**Table 4**

Choice accuracy, PT share bias and mean absolute response bias against stated choices.

| Model               | Singapore ( $n = 332$ ) |                 |               | Shanghai ( $n = 321$ ) |                 |               |
|---------------------|-------------------------|-----------------|---------------|------------------------|-----------------|---------------|
|                     | Acc. (%)                | PT bias (pp)    | Mean $ B_c $  | Acc. (%)               | PT bias (pp)    | Mean $ B_c $  |
| SA-Student          | 66.9                    | -30.7           | 4.0           | 80.5                   | -16.8           | 5.8           |
| MNL-S               | 71.1                    | -3.8            | 11.1          | 81.8                   | -5.4            | 4.1           |
| Soft KL             | $72.9 \pm 0.9$          | $-17.7 \pm 3.3$ | $8.2 \pm 3.3$ | $78.5 \pm 1.7$         | $-12.3 \pm 7.9$ | $7.6 \pm 0.9$ |
| CE+KL               | $69.7 \pm 0.3$          | $-23.5 \pm 3.5$ | $6.9 \pm 1.7$ | $77.1 \pm 3.0$         | $-15.0 \pm 6.8$ | $9.1 \pm 1.9$ |
| Signed response     | $73.1 \pm 1.2$          | $-16.0 \pm 3.5$ | $8.6 \pm 3.0$ | $79.8 \pm 1.1$         | $-10.7 \pm 6.2$ | $7.7 \pm 1.8$ |
| Direction+magnitude | $71.8 \pm 3.1$          | $-16.7 \pm 6.2$ | $8.8 \pm 2.0$ | $80.3 \pm 0.5$         | $-10.4 \pm 5.9$ | $7.5 \pm 1.0$ |

All 3,320 Singapore and 3,208 Shanghai explicit choices are scored, including eight and two choices of modes the model treats as unavailable. PT bias is the mean predicted PT probability minus the stated PT share. Mean  $|B_c|$  is computed within each fitted seed by taking the absolute bias for each contrast and averaging over the five Singapore or eight Shanghai contrasts; these seed-level metrics are then averaged. Figure 4 instead plots signed biases averaged over seeds, whose absolute values need not equal the table metric. Both are in percentage points. Neural entries are means  $\pm$  SD over three training seeds. Always predicting PT gives accuracies of 66.57% and 71.98%. Further metrics are given in Supplementary Table S15.

minutes, access walking from three to twelve minutes and transfers from zero to two (Supplementary Table S2). The reversal thus concerns this combined access burden rather than any single walking-time coefficient. The model with the smallest Teacher error is, for this intervention, the only one that predicts the wrong direction. Conversely, CE+KL, which has the largest Teacher-response error, comes closest to the stated accessibility response ( $-23.37$  against  $-24.40$  points; Supplementary Table S16).

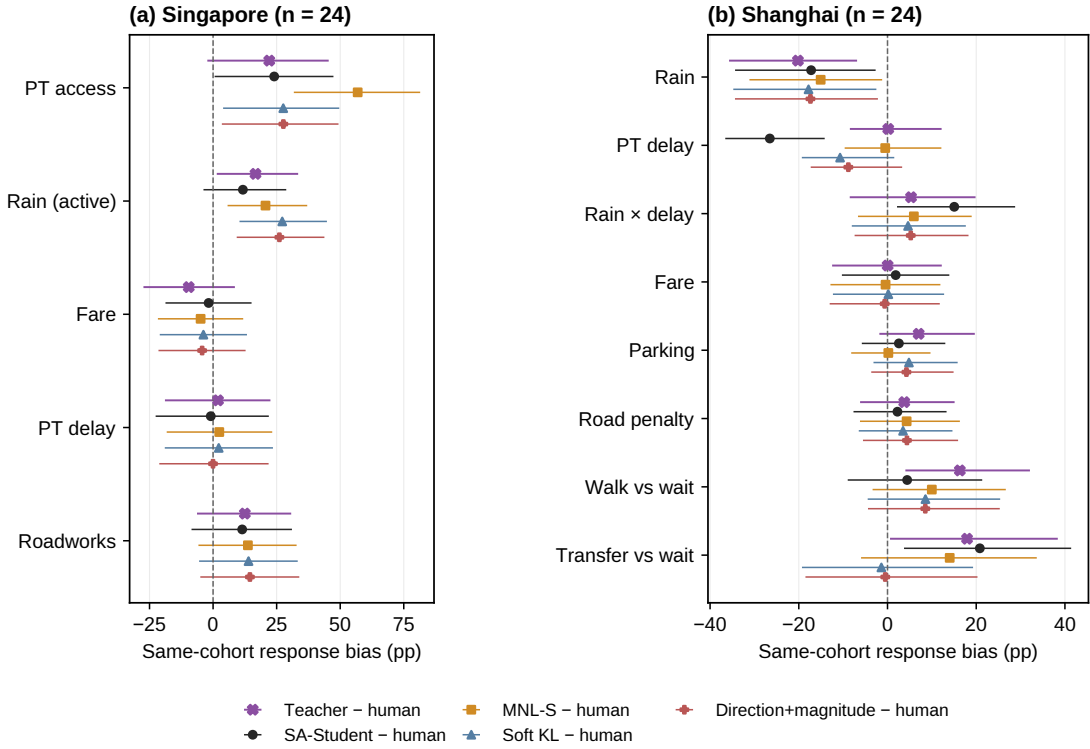
Response bias and level bias also rank the models differently (Table 4). SA-Student underpredicts the PT share by 30.67 points in Singapore and 16.84 in Shanghai, whereas MNL-S narrows these gaps to 3.77 and 5.36 points. In Singapore, however, MNL-S has the larger mean absolute response bias, 11.1 points against 4.0 for SA-Student. A model can therefore reproduce how many people use PT while misrepresenting how that number changes, and the reverse also occurs. Accuracy is a weak guide in either case. SA-Student's accuracy of 66.93% in Singapore barely exceeds the 66.57% obtained by always predicting PT, while in Shanghai it reaches 80.46%.

Numerical inputs affect agreement as well, though not mainly through respondent-specific matching. In Shanghai, replacing reference values with network-derived inputs raises accuracy by 2.80 percentage points, and the gain remains 2.56 points after assigning fresh origin–destination pairs and 2.52 points after exchanging complete supply profiles between respondents. Most of the improvement therefore comes from a general shift in the numerical inputs rather than from matching them to individual respondents (Supplementary Section S3.4).

### 5.3. Discrepancies relative to a contemporary API reference

The matched 24-person subsets describe two components of discrepancy under the reference inputs (Figure 5). They compare current predictions and do not identify the historical origin of the differences. The first component is the contemporary Teacher–human discrepancy. Under poorer accessibility, the selected Singapore respondents reduce PT choice by 37.50 points, while the contemporary Teacher predicts a reduction of 15.40 points and SA-Student 13.46. The attenuation is also observed in the contemporary Teacher, and SA-Student differs from it by +1.94 points (nominal 95% interval  $[-3.40, 7.26]$ ).

The second component is the Student–Teacher discrepancy. For the encoded Shanghai delay task, the selected respondents and the Teacher agree closely, at  $-8.33$  and  $-8.17$  points, but SA-Student predicts  $-34.87$ . Its difference from the Teacher is  $-26.70$  points, with a family-adjusted interval of  $[-36.95, -15.42]$ . The other Students respond less strongly, at  $-19.03$  for soft KL,  $-17.17$  for direction+magnitude and  $-8.85$  for MNL-S, so SA-Student shows the largest discrepancy in this comparison. This pattern does not isolate its training history from changes in the Teacher service.



**Figure 5:** Contemporary API reference and Student response differences on the matched 24-person subsets. The Teacher label denotes the current API reference, not a verified reconstruction of the training backend. Points show the change in model probability minus the change in stated-choice share, and bars show nominal 95% respondent-bootstrap intervals. Teacher means average three elicitations per state, and controlled neural means average three fitted seeds. Paired intervals between Student and Teacher are given in Table 5 and the supplement.

The two terms of Eq. 13 can also offset each other (Table 5). In the Shanghai walking-versus-waiting contrast, the Teacher exceeds the human response by +16.31 points and SA-Student falls 11.88 points below the Teacher, leaving a total bias of only +4.43 points. A small total bias can therefore conceal two larger, opposing discrepancies. For this reason the decomposition is more informative than the total bias for distinguishing the two observed discrepancies, although the Student–Teacher term may also include changes in the Teacher since training.

**Table 5**

Decomposition of signed response differences among stated choices, the contemporary Teacher and SA-Student.

| Contrast                  | Teacher – human     | SA-Student – Teacher | SA-Student – human  |
|---------------------------|---------------------|----------------------|---------------------|
| <i>Singapore (n = 24)</i> |                     |                      |                     |
| Poorer PT access          | +22.1 [−9.9, 52.1]  | +1.9 [−4.9, 9.1]     | +24.0 [−7.2, 54.3]  |
| Rain (active modes)       | +16.7 [−2.4, 38.4]  | −5.0 [−11.4, 1.1]    | +11.7 [−8.0, 33.9]  |
| PT fare                   | −9.6 [−33.4, 14.8]  | +7.8 [4.1, 12.4]     | −1.8 [−26.2, 22.6]  |
| PT delay                  | +1.9 [−25.2, 28.9]  | −2.9 [−12.3, 5.8]    | −0.9 [−28.8, 29.7]  |
| Road penalty              | +12.5 [−12.0, 36.3] | −1.0 [−5.8, 4.2]     | +11.4 [−14.8, 37.0] |
| <i>Shanghai (n = 24)</i>  |                     |                      |                     |
| Rain                      | −20.2 [−41.6, −2.5] | +3.0 [−2.9, 8.0]     | −17.2 [−41.5, 2.3]  |
| PT delay                  | +0.2 [−10.2, 18.0]  | −26.7 [−36.9, −15.4] | −26.5 [−39.3, −8.3] |
| Rain × delay              | +5.3 [−13.9, 26.1]  | +9.7 [3.3, 16.2]     | +15.0 [−2.5, 34.7]  |
| PT fare                   | +0.0 [−16.8, 16.7]  | +1.8 [0.3, 3.2]      | +1.8 [−14.6, 18.3]  |
| Parking                   | +7.0 [−3.0, 24.6]   | −4.5 [−11.0, −0.3]   | +2.6 [−8.1, 17.7]   |
| Road penalty              | +3.8 [−10.3, 20.1]  | −1.5 [−4.7, 0.7]     | +2.2 [−11.6, 18.3]  |
| Walk versus wait          | +16.3 [0.2, 39.6]   | −11.9 [−14.2, −9.1]  | +4.4 [−13.0, 29.2]  |
| Transfer versus wait      | +17.9 [−7.4, 44.4]  | +2.9 [0.4, 5.2]      | +20.8 [−4.0, 46.8]  |

Entries are percentage points with Bonferroni-adjusted respondent-bootstrap intervals, computed within each city over five Singapore or eight Shanghai contrasts. All terms use the same respondents, the same fitted SA-Student and Teacher means over three elicitations. Point estimates are rounded to one decimal place. The SA-Student – Teacher term may include changes in the Teacher since training.

#### 5.4. Response transmission through assignment and execution

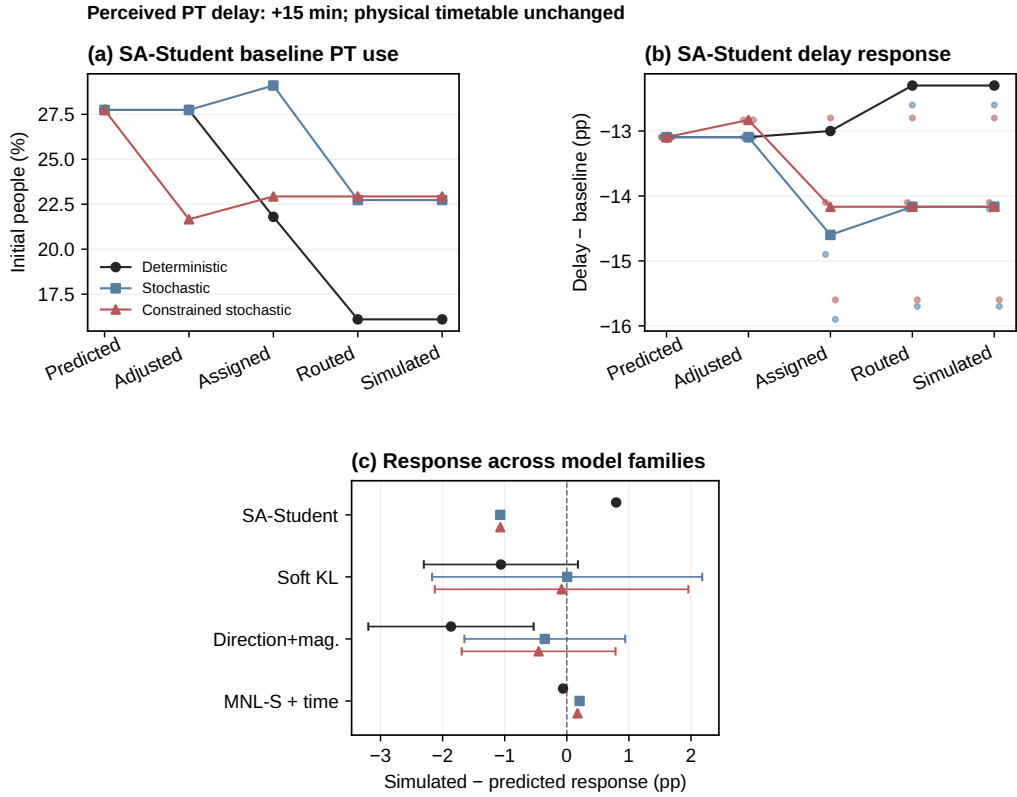
For SA-Student in Helsinki, the predicted response to perceived PT delay is −13.10 percentage points. Deterministic assignment gives −13.00 points, and routed modes and simulated PT use both give −12.30. The resulting gap of +0.80 points has a paired-person 95% interval of [−0.77, 2.38]. The level of PT use behaves very differently. At baseline, mean PT probability is 27.74%, 218 of the 1,000 travelers are assigned PT, and 161 board PT in the simulation, so 73.85% of assigned PT users board before reaching the outbound destination. PT use thus falls by more than eleven points between prediction and simulation, while the delay response changes by less than one point. The distinction between levels and responses that separates static from response fidelity thus reappears at the network stage.

Stochastic assignment represents the predicted distribution in expectation, but a finite population still varies from draw to draw. For SA-Student, ordinary and feasibility-constrained sampling both give a mean simulated response of −14.17 points, with assignment SDs of 1.55 and 1.40 points. Feasibility adjustment changes the probability response to −12.83 points and prevents PT assignments for which the specified search finds no route. It makes the assigned demand routable, but it does not bring the simulated response closer to the original prediction in this population. Routability and response preservation are separate properties of an assignment rule. Eq. 9 further separates their contributions: for the feasibility-constrained SA-Student result, the reported mean responses imply a +0.27-point shift from prediction to adjusted probabilities and a −1.34-point shift from adjusted probabilities to simulation, summing to the −1.07-point total gap. These differences are calculated from the reported rounded means; their component uncertainties are not estimated separately here.

MNL-S with neural timing predicts −6.84 points and yields −6.90 under deterministic assignment (Table 6). Its stochastic and feasibility-constrained means are −6.63 and −6.67. For a given assignment seed, PT use is identical across its three independently fitted timing predictors, so timing variability does not alter this model's simulated mode response.

Signed averages can hide opposing errors. The stochastic-assignment gap of soft KL averages approximately zero across fitted models, but the model-specific means range from −2.17 to +2.18 points. When assignment repetitions are averaged within each fitted model before taking absolute values, the mean absolute gap is 1.45 points for soft KL and 0.91 for direction+magnitude (Supplementary Table S28). In this summary, the objective that preserves Teacher responses better also reaches the network with smaller model-specific gaps, a difference that the signed mean conceals.

Departure timing affects routing even when feasibility does not change. Applying SA-Student's predicted departure adjustment alters the selected itinerary in 279 baseline and 313 delay cases while leaving binary PT feasibility unchanged.



**Figure 6:** Response preservation for a fixed population of 1,000 Helsinki travelers under a 15-minute perceived PT delay, with the physical timetable unchanged. (a,b) Baseline PT shares and delay responses of SA-Student across predicted probabilities, feasibility-adjusted probabilities (Adjusted), assigned modes, routed modes and simulated PT use. Small points in (b) show individual assignment seeds. (c) Predicted-to-simulated gaps by model family. Error bars show training-seed SD after averaging assignment repetitions within each model. Simulated PT use is defined by boarding and does not require journey completion.

The timing output therefore shapes the schedule that travelers meet in the simulation, an effect that a feasibility check alone would not reveal.

Journey times depend on the representation of the network as well as on mode counts. In the Shanghai speed experiment, PT counts stay at 224 at baseline and 134 under delay, and all 321 people complete their journeys. The paired change in outbound time is nevertheless  $-4.44$  minutes with network-derived active-mode speeds and  $+14.04$  minutes with mode-specific speed caps for walking and cycling, the latter with an interval of  $[11.18, 17.05]$  minutes. Identical demand plans can thus imply time changes of opposite sign (Supplementary Section S5.2).

**Table 6**

Predicted and simulated PT responses in the fixed 1,000-person Helsinki population.

| Model               | Assignment    | Predicted (pp)   | Simulated (pp)   | Gap (pp)         | Assignment SD |
|---------------------|---------------|------------------|------------------|------------------|---------------|
| SA-Student          | Deterministic | -13.10           | -12.30           | 0.80             | –             |
| SA-Student          | Stochastic    | -13.10           | -14.17           | -1.07            | 1.55          |
| SA-Student          | Constrained   | -13.10           | -14.17           | -1.07            | 1.40          |
| Soft KL             | Deterministic | $-6.97 \pm 4.92$ | $-8.03 \pm 5.13$ | $-1.06 \pm 1.24$ | –             |
| Soft KL             | Stochastic    | $-6.97 \pm 4.92$ | $-6.97 \pm 3.32$ | $0.01 \pm 2.18$  | 0.27          |
| Soft KL             | Constrained   | $-6.97 \pm 4.92$ | $-7.06 \pm 3.41$ | $-0.08 \pm 2.04$ | 0.28          |
| Direction+magnitude | Deterministic | $-6.77 \pm 4.96$ | $-8.63 \pm 4.76$ | $-1.87 \pm 1.33$ | –             |
| Direction+magnitude | Stochastic    | $-6.77 \pm 4.96$ | $-7.12 \pm 3.97$ | $-0.35 \pm 1.30$ | 0.35          |
| Direction+magnitude | Constrained   | $-6.77 \pm 4.96$ | $-7.22 \pm 4.06$ | $-0.45 \pm 1.24$ | 0.32          |
| MNL-S + neural time | Deterministic | $-6.84 \pm 0.00$ | $-6.90 \pm 0.00$ | $-0.06 \pm 0.00$ | –             |
| MNL-S + neural time | Stochastic    | $-6.84 \pm 0.00$ | $-6.63 \pm 0.00$ | $0.21 \pm 0.00$  | 1.10          |
| MNL-S + neural time | Constrained   | $-6.84 \pm 0.00$ | $-6.67 \pm 0.00$ | $0.17 \pm 0.00$  | 1.21          |

Responses are changes in PT use from baseline to perceived delay over the same 1,000 people, with the physical timetable unchanged. Gap is the simulated minus the predicted response. Constrained denotes feasibility-constrained stochastic assignment. Values average assignment repetitions within each fitted model and then across models, and  $\pm$  gives the SD across three training seeds. SA-Student is a single fitted model. Assignment SD is the mean within-model SD across assignment repetitions. Intermediate stages and absolute-gap summaries are given in Supplementary Tables S27–S28.

### 5.5. Computational implications for scenario analysis

In the measured post-training builds, local behavioral prediction accounts for less than one percent of scenario-construction time. Constructing a scenario for 10,000 Singapore agents takes 51.1 minutes, of which behavioral prediction accounts for 10.3 seconds. For 50,000 agents the corresponding times are 231.8 minutes and 48.8 seconds, so inference contributes approximately 0.34% and 0.35% of construction time. In a model-only benchmark that excludes feature encoding and routing, MNL-S with neural timing requires  $1.68 \pm 0.33$  microseconds per state, compared with 1.78–2.16 microseconds for the joint neural objectives.

The dominant cost lies in network and timetable calculations. Reusing those that do not change between scenarios reduces construction time from 3,223.6 to 469.5 seconds in Singapore and from 10,491.8 to 103.2 seconds in Helsinki, with mode and origin–destination assignments unchanged. Each city was measured under one reuse condition, and concurrent workload limits the precision of these timings. The differences among compact Students are negligible by comparison.

The archived usage records also permit a monetary estimate without new Teacher calls (Table 7). Across the retained archives, 13,391 deduplicated completions with usage consumed 17.157 million input and 50.418 million output tokens. Repricing them at the official V4 Pro rates gives US\$102.71–205.42; the final same-task reference accounts for US\$7.08–14.16. These are rate-card scenarios for observed usage, not provider invoices or a complete lifecycle total. In particular, 576 earlier completions lack usage, and the retained archives do not cover every failed attempt or acquisition source. Reasoning tokens are already included in output tokens.

For repeated deployment, let  $C_0$  denote acquisition, retry, training and necessary adaptation costs, and let  $c_L$  and  $c_S$  be the monetary costs of evaluating the same scenario directly with the LLM and with the Student. The break-even workload is

$$N^* = \frac{C_0}{c_L - c_S}, \quad c_L > c_S. \quad (14)$$

Shared feature, supply and routing work must be counted consistently in both pipelines, with matching caching and concurrency assumptions. Existing inference, construction and token records support the components above, but the missing acquisition charges and local-compute tariff prevent a measured lifecycle break-even claim. Wall-clock and resource break-even require separate calculations in their own units; successful-request latency cannot replace elapsed campaign time (Supplementary Section S6).

**Table 7**

Retained API usage repriced at the official DeepSeek V4 Pro rates checked on 25 September 2026. Costs are estimates in USD, not invoices.

| Acquisition group            | Metered calls | Input (M) | Output (M) | USD range     |
|------------------------------|---------------|-----------|------------|---------------|
| Primary supervision archives | 8,681         | 10.050    | 32.614     | 65.83–131.66  |
| Final same-task reference    | 1,440         | 2.324     | 3.255      | 7.08–14.16    |
| Earlier same-task campaign   | 1,475         | 1.969     | 5.575      | 11.58–23.15   |
| Neutral-metadata pilot       | 120           | 0.160     | 0.239      | 0.52–1.03     |
| Ownership pilot              | 120           | 0.160     | 0.235      | 0.52–1.03     |
| Latency benchmark            | 87            | 0.117     | 0.353      | 0.76–1.51     |
| Other historical development | 1,468         | 2.377     | 8.149      | 16.43–32.86   |
| Retained metered total       | 13,391        | 17.157    | 50.418     | 102.71–205.42 |

Completion IDs are deduplicated across archives. Input costs use the recorded cache-hit/miss split; output includes reasoning tokens. The range reprices the same usage entirely at off-peak or peak rates, rather than reconstructing historical deductions. The primary archives contain the legacy, joint and corrected accessibility acquisition pools, including labels outside the final benchmark. Other development includes deprecated S8. A further 576 retained completions lack usage and are excluded from the monetary sum, as are unmetered failures; their cost is unknown. The earlier same-task campaign includes 35 responses rejected by parsing but with recorded token usage. This is neither the marginal cost of one fitted Student nor a full project invoice.

## 6. Discussion

### 6.1. Response fidelity is a separate model-selection criterion

The central result is not that a particular loss or architecture wins every comparison. It is that Teacher-response fidelity, agreement with stated choices and execution consistency rank models differently. In the controlled benchmark, explicit response weighting improves finite-contrast error, but changing the choice specification produces a larger gain. The signed and direction+magnitude objectives are not clearly separated by the paired interval, and neither improves interaction error over soft KL. The evidence therefore supports evaluating and weighting responses, without establishing that the split sign–magnitude construction is essential.

This interpretation follows the information structure of the experiment. The same endpoint targets are available to every controlled model, and Eq. 7 connects their errors to response error. Paired supervision reallocates optimization effort across those errors rather than revealing additional Teacher behavior. Results across the six held-out personas and the available Teacher reaggregations support the within-benchmark comparison. Transfer to unseen intervention families, intensities or Teachers remains a separate empirical question.

MNL-S’s advantage makes model specification an important control rather than a secondary baseline. Its linear utilities provide inspectable coefficients, and separate neural timing retains most of the timing accuracy of joint models. This combination supports modular estimation and later recalibration. It does not establish that travelers follow linear utilities or that every coefficient has an economically valid interpretation: the choice target is an LLM judgment, and smoothness or regularization could also explain the gain. Behavioral interpretation requires checking identified utility differences, relevant signs and implied sensitivities, not merely naming the model logit.

### 6.2. Human agreement requires an independent empirical test

Teacher fidelity is not a proxy for agreement with people. MNL-S reverses the stated Singapore accessibility response, whereas CE+KL, the least faithful controlled objective, comes closest to that contrast. Baseline levels and responses also separate: SA-Student understates Singapore PT use by about 31 percentage points yet has the smallest mean absolute response bias there; MNL-S comes within four points of the PT share but misses important changes. These observations extend the distinction between predictive and behavioral criteria (Wang et al., 2020; Martín-Baos et al., 2023) to models compressed from LLM judgments.

The contemporary API reference helps identify where follow-up tests are needed, without assigning historical causation. For Singapore accessibility, it also attenuates the stated response, so bringing SA-Student closer to that reference would leave substantial human disagreement. For Shanghai delay under the reference encoding, the current model is close to the human point estimate while SA-Student responds much more strongly. The second pattern motivates a controlled check of Student specification and input encoding; it does not establish that compression caused the discrepancy. The Shanghai walking-versus-waiting result additionally illustrates cancellation: a small total bias can

combine larger opposing Student–reference and reference–human terms.

This distinction suggests two different development tasks. Compression should be evaluated against frozen Teacher targets. Empirical alignment should be fitted and selected using separate human data, then tested on people and tasks not used for that alignment, as motivated by recent travel-choice and travel-satisfaction research (Liu et al., 2026c; Xu and Wang, 2026). The surveys in the present study were reserved for evaluation, so the observed differences diagnose calibration needs rather than demonstrate a calibrated traveler model.

### 6.3. Routability, response transmission and practical use

The Helsinki experiment isolates what happens after behavioral prediction. For SA-Student, baseline PT use falls by more than eleven points between prediction and boarding while the deterministic delay response changes by less than one point. A small response gap can thus coexist with a substantial level gap. Feasibility-constrained assignment removes PT choices for which the specified search finds no connection, yet does not reduce the response gap for this population. Routability and preservation of a prescribed behavioral distribution are different objectives.

The stagewise accounting makes these differences auditable. All shares use the same initial population; predicted and adjusted probabilities remain distinct from assigned modes, routed modes and boarding events. Signed and absolute gaps are both needed because cancellation across fitted models can conceal errors. Timing outputs also warrant separate evaluation: they change itineraries even when binary feasibility remains unchanged. The additional speed experiment shows why a stable mode count alone cannot validate a journey-time outcome.

For implementation, the relevant efficiency result concerns repeated use after training. Behavioral inference consumes approximately 0.35% of the measured larger scenario builds, shifting attention to supply calculation and reuse rather than marginal differences between compact Students. The retained token ledger quantifies an additional acquisition-cost component, but it does not yet determine the lifecycle break-even workload: unmetered attempts and local-compute costs remain outside the estimate in Table 7.

### 6.4. Scope of inference and next tests

Three boundaries define the conclusions. First, the compression comparison uses one archived Teacher and six held-out synthetic personas. Teacher reaggregation and seed variation test sensitivity within that design; they do not substitute for new personas, unseen interventions or another frozen Teacher. Second, the human evidence concerns stated responses in two convenience samples under documented encodings. The cities used different task formats, Shanghai’s qualitative tasks require numerical reference values, and the existing sensitivity analysis holds its delay at 15 minutes. Explicit currency, alternative-set interpretation and randomized, numerically matched tasks are priorities for further validation. Third, Helsinki holds physical supply and behavioral predictions fixed to isolate execution effects. Physical service changes and feedback from experienced conditions require separate experiments before extending the conclusions to equilibrium or operational disruption forecasting.

These boundaries define a usable validation sequence rather than a single pass/fail imitation score: specify the intervention and output; test compression against archived targets; compare the corresponding human response; and audit the outcome after assignment and execution. A failure at one level identifies the next model component or evidence source to examine. Supplementary Section S7 provides the versioned material inventory and distinguishes reproducible target-based evaluation from live-API and restricted-data dependencies.

## 7. Conclusion

This study evaluates LLM-derived traveler models through finite responses rather than baseline imitation alone. On the controlled benchmark, response-aware supervision improves Teacher fidelity and a logit choice model with independent neural timing performs better still on response error. That ranking does not carry over to the stated-choice samples: the Teacher-best Student reverses a human accessibility response. In the controlled Helsinki execution experiment, improved routability likewise does not ensure closer transmission of the predicted response.

The practical contribution is a response-centric validation procedure linking compression, empirical comparison and network execution. Each level answers a different question, and success at one cannot stand in for evidence at another. A traveler model intended for a specified intervention should therefore be evaluated on its response at all relevant levels, with baseline demand and computational cost reported separately. The current evidence establishes these distinctions within the tested Teacher, samples and execution setting; broader forecasting claims require the corresponding out-of-setting tests.

## Data and code availability

Code, prepared synthetic benchmark states and targets, split manifests, selected Student checkpoints, evaluation scripts and compact result tables are included in the research repository, [https://github.com/AnXu-ITS/DeepSeek\\_Traveler\\_Behavior\\_Distillation\\_Research](https://github.com/AnXu-ITS/DeepSeek_Traveler_Behavior_Distillation_Research). The experimental artifacts are pinned to the 24 September 2026 repository edition at commit 4e67286; the 25 September editorial revision adds aggregate cost accounting and corrected service and consent statements without new experiments. The complete base identifier and material inventory are given in Supplementary Section S7. Archived targets permit controlled-model evaluation without new Teacher calls. The repository remains access-controlled; requests should be directed to the corresponding author. Participant-level survey records and participant-linked API payloads are retained separately, subject to privacy and applicable permissions. The repository includes Helsinki run-level summaries and analysis code; fresh simulation additionally requires the documented supply files and MATSim environment. Supplementary Section S7 specifies the coverage of each reproduction task.

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